

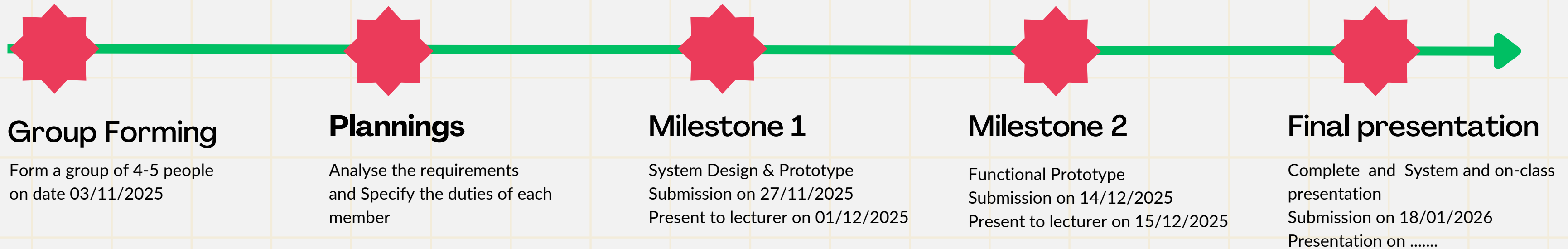


REAL-TIME TRAFFIC SIMULATION WITH SUMO PLANNING & TRACKING

Slides made by **Duc Thien Nguyen**

PROJECT TIMELINE

TEAM MEMBER
MARCO
DUC THIEN
HOANG BACH
MINH PHU



OUR MAIN ROLES

	ROLE	RESPONSIBILITIES
MARKO	SUMO INTERGRATION TEAM LEADER	SUMO CONNECTION VIA LIBTRACI API , WRAPPER
DUC THIEN	DOCUMENTATION AND TESTING	user manual/screenshots, UML diagrams, milestone PDF assembly, declarations of authorship.
HOANG BACH	DATA ANALYSIS	Traffic statistics computation, real-time charts, CSV and PDF report generation, file and stream handling
MINH PHU	CORE SIMULATION	Simulation loop implementation, SUMO stepping logic, thread management, synchronization between simulation and GUI

MILESTONE ONE

Due on : 27/11/2025

Project: New campaign

Task	Status	Priority	Done Date	Owner	Lecturer's Comment
SUMO connection demo and libtraci wrapper	✓ Completed ▾	● High ▾	27 Nov 2025	Marko	Add more features in milestone 2
GUI Mockups	✓ Completed ▾	● Medium ▾	27 Nov 2025	Lazar	Make GUI and Visualize
Sumo networking creating and testing	✓ Completed ▾	● High ▾	27 Nov 2025	Hoang Bach	Absent in Milestion 1
Sumo networking creating and testing	✓ Completed ▾	● High ▾	27 Nov 2025	Minh Phu	Absent in Milestion 1
Report and architecture diagram	✓ Completed ▾	● High ▾	27 Nov 2025	Duc Thien	Fix the diagram and make the timeline

MILESTONE TWO

Due on : 14/12/2025

Project: New campaign

Task	Status	Priority	Done Date	Owner	Notes
Sumo Wrapper, check API key	✔ Completed ▾	● Medium ▾	14 Dec 2025	Marko	
GUI (Center control)	🕒 Not started ▾	● High ▾	📅	Lazar	Withdrawed
SUMO Stress Test	🕒 Not started ▾	● High ▾	📅	Hoang Bach	
Application (Map visualization, Vehicle injection. traffic light	✔ Completed ▾	● High ▾	09 Dec 2025	Minh Phu	
User Guide Draft, Testing, Report	✔ Completed ▾	● Low ▾	14 Dec 2025	Duc Thien	GUI draft (using Swing , look at JavaFX later)

FINAL PREPARATION

Due on : 14/12/2025

Project: New campaign

Task	Status	Priority	Done Date	Owner	Notes
Filtering vehicles by color and speed	✓ Completed ▾	● High ▾	17 Jan 2026	Marko	
	🕒 Not started ▾	● High ▾	📅	Lazar	Withdrawed
File export, and statistic with chart and summary	✓ Completed ▾	● High ▾	18 Jan 2026	Hoang Bach	
Map Rendering	✓ Completed ▾	● High ▾	17 Jan 2026	Minh Phu	
Traffic light management and vehicle injection on specific edge	✓ Completed ▾	● High ▾	18 Jan 2026	Duc Thien	